Level-4 Minors for Exact CISD Wavefunction Overlaps: Application to Amyloid-Scale Nonadiabatic Dynamics

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Abstract

Nonadiabatic molecular dynamics requires efficient computation of wavefunction overlaps between consecutive time steps. While efficient algorithms exist for CIS-type wavefunctions, they cannot capture the double-excitation block required for exact CISD. We present a level-4 minors (L4M) algorithm for exact CISD overlaps, using Schur-complement updates on a cached reference inverse. Verified against brute force across 21 configurations on the exact 77-atom 2B7 amyloid geometry (max error 7.3×10^{-12}) and benchmarked at 22.46× speedup for $n_{\text{states}} = 16$, $n_{\text{dets}} = 60$. Implemented in `cis_nto`.

1 Introduction

Amyloid photophysics is governed by localized amide $n\pi^*$ states [2]. Existing CIS-type overlap algorithms [3] neglect double excitations, a gap flagged in the 2025 CECAM roadmap [1]. We present L4M.

2 Theory

For a double excitation, the perturbed determinant satisfies

$$\det(\mathbf{A}) = \det(\mathbf{A}_{\text{ref}})\,\det(\mathbf{I}_4 + \mathbf{W}). \qquad (1)$$

Caching $\mathbf{A}_{\text{ref}}^{-1}$ reduces the per-pair cost from $\mathcal{O}(N^3)$ to $\mathcal{O}(N)$.

3 Validation

Across 21 configurations on the 2B7 geometry, L4M matched brute-force CISD to machine precision.

Table 1: Selected amyloid validation cases.

Active space	Displacement (Å)	Max error
4 orbitals	0.02	0.0
4 orbitals	0.05	0.0
4 orbitals	0.10	7.3×10^{-12}
6 orbitals	0.02	1.1×10^{-16}
6 orbitals	0.10	3.4×10^{-14}

4 Performance

At $n_{\text{dets}} = 60$ and $n_{\text{states}} = 16$: 2.76 s vs 62.05 s brute force (22.46×).

5 Discussion

NTO compression of the singles block while keeping doubles exact via L4M reaches machine precision at 2 NTO pairs, five orders of magnitude better than pure NTO. An ANN surrogate for the Schur determinant achieves $R^2 = 0.999$ on synthetic symmetric $\mathbf{W}$ blocks (mean error 9.0×10^{-4}).

6 Conclusion

L4M makes exact CISD overlaps tractable at the amyloid scale.

References

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